Infrastructure Maintenance Manual

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CCH® Tagetik

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Introduction

The purpose of this manual is to provide general maintenance and backup guidelines for the application after the core application has been successfully installed and correctly configured.

The application is deployed on a three tiered JEE enterprise application which consists of a JEE compliant application server (AS) tier, a relational database (DB) server tier, and a client tier. Each server tier can be installed in on virtual, physical, cloud or container infrastructure. The application's flexible architecture allows it be deployed on varied infrastructures with different AS and DB infrastructure software and resources and integrated with various network configurations, load balancers and other third party software packages.

This manual attempts only to cover the most essential principles and procedure of maintenance and backup of the AS and DB tiers with some mention of client tier maintenance. Other third party external software packages may require different or additional maintenance tasks. Please note that customer information management and security policies should be consulted in conjunction with this manual before carrying out maintenance tasks.

Requirements

Readers of this manual should have good understanding of the architecture of the application and the infrastructure that hosts the core application and more detailed knowledge of the management and maintenance of at least of one of the three tiers mentioned above. In addition, it is helpful if readers have some experience in the daily operations of the application from the technical perspective, i.e. know how to start/stop the application, have installed or upgraded the application or its .Net Add-ins for Microsoft Office, and/or maintained its databases.

Many of these procedures and tasks require specific access and privileges to carry out. Users may also require application administrative user credentials such as username and password credentials to access application functions.

Additional Information

Additional technical information can be found in the technical and user manuals available on the CCH Tagetik Community (http://community.tagetik.com). If you have used the installation bundle containing WildFly from the CCH Tagetik Community, there may also be a documents directory which contains versions of the technical manuals. N.B. Please note that the contents of the documents directory may not be the most up-to-date. Registered customers can find all of these documents for download from the CCH Tagetik Community.

ATTENTION: The maintenance and backup guidelines in this manual may require coordination with the users of the application in order to avoid lost data or diminished application performance and response times.

Please contact CCH Tagetik concerning any questions or comments related to the contents of this manual.

Application Server Tier

This section covers the general maintenance and backup of the AS tier regardless of the vendor software selected. This tier of the infrastructure generally requires less management than the DB tier unless there are resource contention issues.

N.B. Check vendor documentation for vendor specific maintenance tasks and requirements.

Logs and Logging

If the AS has been configured according to the installation manual provided by Tagetik, the application will produce system level logs in text format of the events which occur in the application including login attempts, logouts, application imports, upgrades to the application, and connections with external software such LDAP. N.B. The application also stores entries of these activities and events in a built-in Audit log which stores its entries in the application database.

Depending on the logging configuration implemented, it is suggested that the space occupied by the application server logs be checked periodically. The default server logging configuration will rollover the current day's log file to create a new log file each day. The log directories are generally configured in the installation manual as one of the following locations:

- on Windows systems is usually found in **`<drive letter>:\tagetik\logs`**, and
- on Unix-like or Linux systems in **`/opt/tagetik/logs`**.

Depending on the parameters chosen during installation procedure, the application system log directory may be found in a different location. The parameter for the location of the AS system logs is set in the xml file which is located by the following Java system property, **`-Dlog4j.configuration`**, as in this example for Windows,

```
-Dlog4j.configuration=file:///tagetik/config/log4j/log4j.xml
```

Therefore, it may be necessary to check the settings of this configuration file. Regarding retention of the application log files, for diagnostic purposes we generally suggest retaining two weeks of logs if possible. However, If there are applicable governance compliance regulations, please consult them before modifying, moving or deleting any server log files.

N.B. For older installations of the application, i.e. pre-4.2, the log file naming convention had two files named **`server.log`** and **`memory.log`**.

Each AS instance, regardless of vendor, may also have logging configured. Administrators should check that the logging configuration does not produce too many log files.

For users of the CCH Tagetik configured JBoss or WildFly AS, it may also be helpful to check the **`..\<jboss|wildfly>\standalone\log`** directory for files which may be archived or deleted.

Import/Export Directory

The core application provides a built-in tool for importing and exporting data via the web interface. The application requires a directory on the application server file system to store the files resulting from the import or export activities and any intermediate files. These files are generally deleted automatically by the

application after a default period of 7 days. The time period can be changed by setting a Java system property in the JVM configuration file,

```
-Dit.grupposervizi.easy.deleteFilesOlderThan=<file age in days>
```

The directory location is generally configured during the installation procedure by using the following system property,

```
-Dit.grupposervizi.easy.impexp.writableFolder=<drive>:\<path>\<impexpdir>
```

which must be a directory which exists and is writable by the AS. In installations utilizing WildFly and configurations provided by CCH Tagetik, this is set in the **`standalone.conf`** or **`standalone.conf.bat`** or equivalent file.

If the import/export directory is occupying excessive storage space, the files can be deleted directly from the file system or via the application web interface in the Import/Export tool by users with access to the Repository, i.e. have administrative roles. We also suggest notifying the application administrator(s) or consultant(s) as some of the files may be necessary for project activities.

Backups and Restores

The AS tier files and configurations do not change often and need backups much less frequently than the DB tier as all the application business data is stored on the DB tier. We suggest backing up the AS tier infrastructure after the installation and after each upgrade.

On the AS tier, a backup can be made by creating an image of the entire server VM. It may also be enough to make copies of the configuration files and the ear file provided by CCH Tagetik. If required, CCH Tagetik can provide the ear file again.

As an example, in installations which rely on the WildFly / JBoss and the standard installation configuration, the configuration files can be found on Windows systems in the **`<drive letter>:\tagetik`** directory and its subdirectories and in the WildFly / JBoss sub-directory **`<drive>\...\<wildfly|jboss>\standalone\configuration`**.

During restoration of the AS tier, the data source configurations and the JVM configurations should be checked before attempting to start the JVM and the application.

Database Tier

In the application, all application and business configuration and data is stored on the database tier which means that it is essential to have frequent backups and to carry out recommended maintenance activities.

Regardless of the RDBMS utilized by the customer, general maintenance of the application databases or schemas consists principally in the following activities,

- Full rebuilding or renewing the indexes to reflect recent changes in the data on a weekly or daily basis depending on the usage profile of the application.
- Recalculating or refreshing the plans or statistics which every RDBMS uses to optimize the execution of queries on a daily basis.

The frequencies of these activities can have an impact on the performance of the application.

Unless the customer has a specific backup creation and retention policy, CCH Tagetik suggests a full database backup every week, daily incremental backups and retaining database backups for at least 2 weeks.

Database or Schema Naming Conventions

If the databases or schemas are configured according to the conventions outlined in the CCH Tagetik technical manuals then the name format is generally,

- TGK_<environment name>_<number>, or
- TCPM_<environment_name>_<number> or

but the names may vary depending on the customer and project requirements and conventions. Each application requires a minimum of two databases, one for the application configuration information and one for the business data. By convention, the database/schema containing application configurations is considered a “repository” and has as part of its name REPOS or REPOSITORY.

COMPATIBLE RDBMS ADDITIONAL DETAILS

The following sub-sections provide additional specific details for some of the RDBMS with which Tagetik is currently compatible. For more details about compatible RDBMS, please consult the CCH Tagetik Compatibility Matrix or the CCH Tagetik Hardware and Software Architecture Overview, both of which are available from the CCH Tagetik Community web portal.

SQL Server

Maintenance

As mentioned earlier, the statistics for each application database should be updated on a daily basis utilizing the syntax “WITH FULLSCAN”. Given the nature of the activities of the application with respect to the databases, the “Auto Update Statistics Option” does not produce optimal query plans reliably. Recalculation of the statistics is suggested for all tables except for temporary tables, i.e. the names of which begin with #.

CCH Tagetik also suggests a rebuild of all the application database indexes.

Backup

Backups can utilize any standard database backup approach, as long as it is compatible with SQL Server, i.e. SQL Server backup/restore, physical copy of data files, any external third party software, etc.

Oracle

Maintenance

As mentioned earlier, the statistics for each Tagetik database should be updated on a daily basis utilizing dbms_stats.gather_schema_stats as recommended by Oracle. A weekly or more frequent rebuild of the indexes is recommended. Recalculation of the statistics is suggested for all tables except for temporary tables, i.e. the names of which begin with #.

CCH Tagetik suggests a minimum of a weekly full rebuild of all the application database indexes.

Backup

Any standard database backup utility, as long as it is compatible with Oracle (exp/imp, expdp/impdp, physical copy of data files, any external agent, etc.) can be used.

PostgreSQL

Maintenance

As mentioned at the beginning of this section, the statistics for each Tagetik database should be updated on a daily basis utilizing the appropriate tools and commands in PostgreSQL. Recalculation of the statistics is suggested for all tables except for temporary tables, i.e. the names of which begin with #.

Backup

Any standard database backup utility, as long as it is compatible with PostgreSQL (pg_dump, pg_dumpall, physical copy of data files, any external software, etc.) can be used.

SAP HANA

Maintenance

As mentioned at the beginning of this section the statistics for each Tagetik database should be updated on a daily basis utilizing the appropriate tools and commands in SAP HANA. Recalculation of the statistics is suggested for all tables except for temporary tables, i.e. the names of which begin with #.

Backup

Any standard database backup utility, as long as it is compatible with SAP HANA can be used.

Client Tier

CCH Tagetik provides .Net Add-ins for the Microsoft Office Suite, i.e. for Excel, Word, and PowerPoint. This section provides some guidelines for maintenance operations on the end user's workstation or where the CCH Tagetik .Net Add-ins may be installed.

Maintenance

It may be necessary to clean the Office document cache occasionally especially if the user downloads reports and/or forms from CCH Tagetik web interface. Using a Microsoft recommended approach is suggested.

CCH Tagetik .Net Add-ins may also have logging configured. The default location for the log files is in the Add-in installation directory, but it may be necessary to access the .Net Add-in configuration to locate the log files. Since the default configuration for logging during installation is to leave logging disabled, excessive .Net Add-in logs are generally a very minor risk.

Backups

The Add-in installation files are easily replaced because they are downloadable from the application web interface. There are some configuration files which are usually located in the Add-in installation directory depending on the workstation configuration. These files include **`appSettings.xml`**, and **`userSettings.ini`**. Optionally, you may find a **`log4net.config`** file if logging was configured for the Add-in(s).

Restoration of Backups and Space

This section covers some additional operations which are not strictly related to maintenance but may be useful to the reader who is maintaining the core application.

Recovery

When restoring the application from backups, please remember that the databases must be restored and accessible before the application server is restored and started. On the AS tier, the application requires access to the repository of application information to successfully start.

Tables and Disk Space

The application stores both Auditing and ETL logs inside its application databases and over time the tables holding this information can cause the Tagetik databases or schemas to grow and occupy a significant amount of disk storage space.

ATTENTION: This data must not be modified in any way without coordination with application administrators and/or consultants.

Auditing data is stored in the repository and holds in all entries of all user or application activities within the application. The Audit log allows the application administrator(s) to track who has accessed the application,

changed, added or deleted data and performed what functions. The Audit data is stored in multiple tables in the Repository. The application provides a tool for managing this data, including deleting or archiving it.

N.B. Archiving audit entries within the application will not address storage issues as the audit entries are moved to read-only tables within the repository.

Attention: CCH Tagetik recommends contacting the application administrator(s) or consultant(s) if there are storage issues related to the Audit log. There may be regulatory compliance issues with handling this data. In addition, to avoid any possible risk of data inconsistency or corruption, it is strongly recommended to use the built-in Audit log management tools.

The ETL logs are stored in each business database and hold all the entries of each ETL task as well as intermediate and temporary data. Some of the tables in the business database or schema include

- MAP_DATI_AQUISITI: contains all the historical data that several routines acquired
- MAP_DATI_TRASFORMATI: contains all the historical data that several routine elaborated from input/acquire data
- MAP_ERRORE%: contains all the logs which refer to ETL.

Attention: As with the Audit logs, CCH Tagetik recommends contacting the application administrator(s) or consultant(s) if there are storage issues related to the ETL tables in a business database or schema. Some of this data may be required for different functions inside the application and/or there may be regulatory compliance issues with handling this data. In addition, to avoid any possible risk of corruption, it is best to use the built-in ETL management tools to handle any modification of this data.

Conclusion

The previous recommendations are not intended to be exhaustive and applicable to all situations. Your particular installation and internal policies may require slightly different tasks or additional tasks. If you have any questions about the above steps, please do not hesitate to contact CCH Tagetik Support or your consultants.

Glossary

The following table provides a short glossary for the terminology used in this manual.

AS	Application Server
DB	Database
RDBMS	Relational Database Management System, i.e. MS SQL Server, Oracle, PostgreSQL
JEE	Java Platform Enterprise Edition
EAR	JEE Enterprise Archive, a file format used in JEE enterprise applications.
JBoss/WildFly	JBoss and WildFly are freely available open source application servers. It can be used to run CCH Tagetik which is a JEE 6 compliant application.

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